Marketing Media Performance Analysis

1. KPI Calculations

To evaluate campaign performance, key marketing KPIs were calculated using the following formulas.

Cost Per Mille (CPM)

Measures the cost of 1000 impressions.

CPM = (Spend / Impressions) * 1000

Interpretation:
Lower CPM means impressions are being generated more cost-efficiently.

Cost Per Click (CPC)

Measures how much is paid for each click.

CPC = Spend / Clicks

Interpretation:
Lower CPC indicates more efficient traffic acquisition.

Cost Per View (CPV)

Measures cost per impression/view.

CPV = Spend / Impressions

Interpretation:
Useful to understand media buying efficiency.

Conversion Rate (CVR)

Measures how efficiently clicks convert to customers.

Conversion Rate = Conversion / Clicks

Interpretation:
Higher conversion rate indicates better campaign targeting or landing page effectiveness.

Return on Ad Spend (ROAS)

Normally calculated as:

ROAS = Revenue / Spend

However, since the dataset does not include revenue, a **proxy metric** was used as Conversions/Spend

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Cost Per Acquisition (CPA) ***(additional KPI)***

CPA measures how much is spent to acquire one conversion.

CPA = Spend / Conversions

Interpretation:
Lower CPA means better marketing efficiency.

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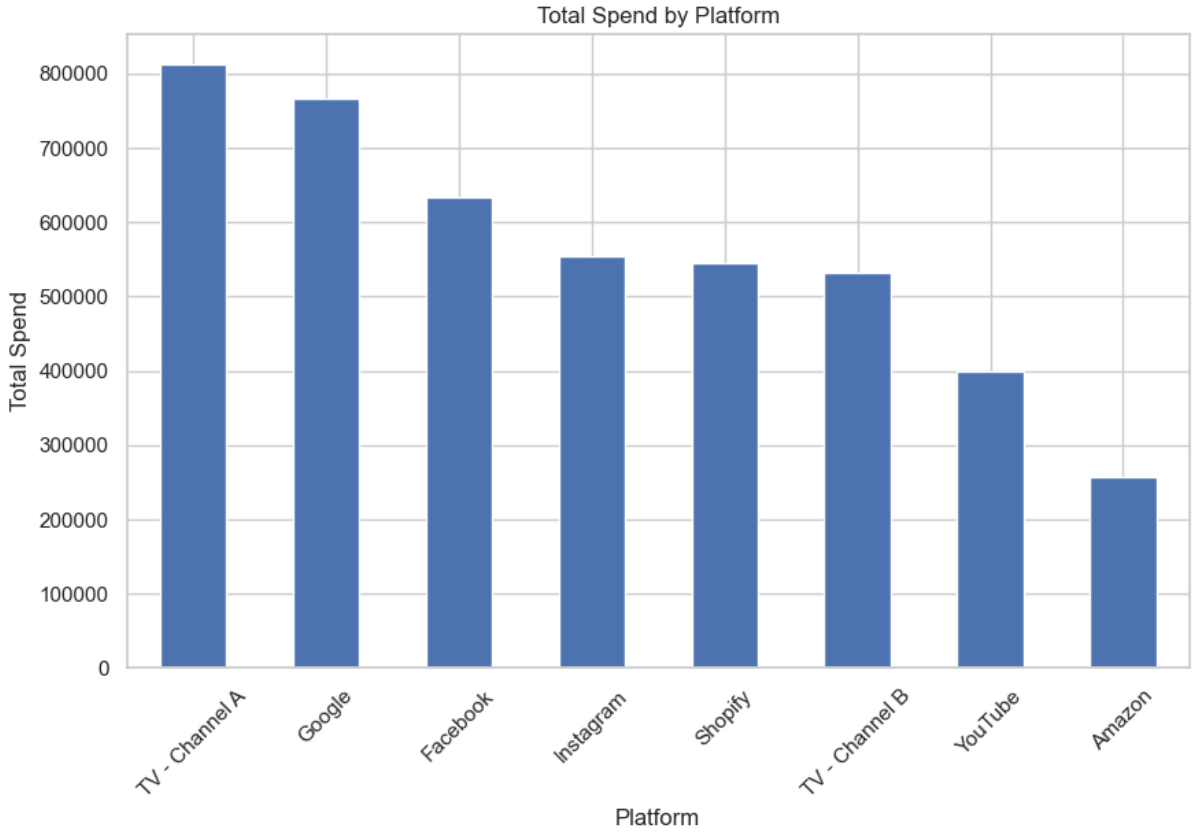
2. Performance Analysis

Campaign performance was analysed by **media channel (platform)** and **campaign audience type**.

Spend Distribution by Platform

Top platforms by spend:

1. TV Channel A – $812,134
2. Google – $766,271
3. Facebook – $634,591
4. Instagram – $553,863

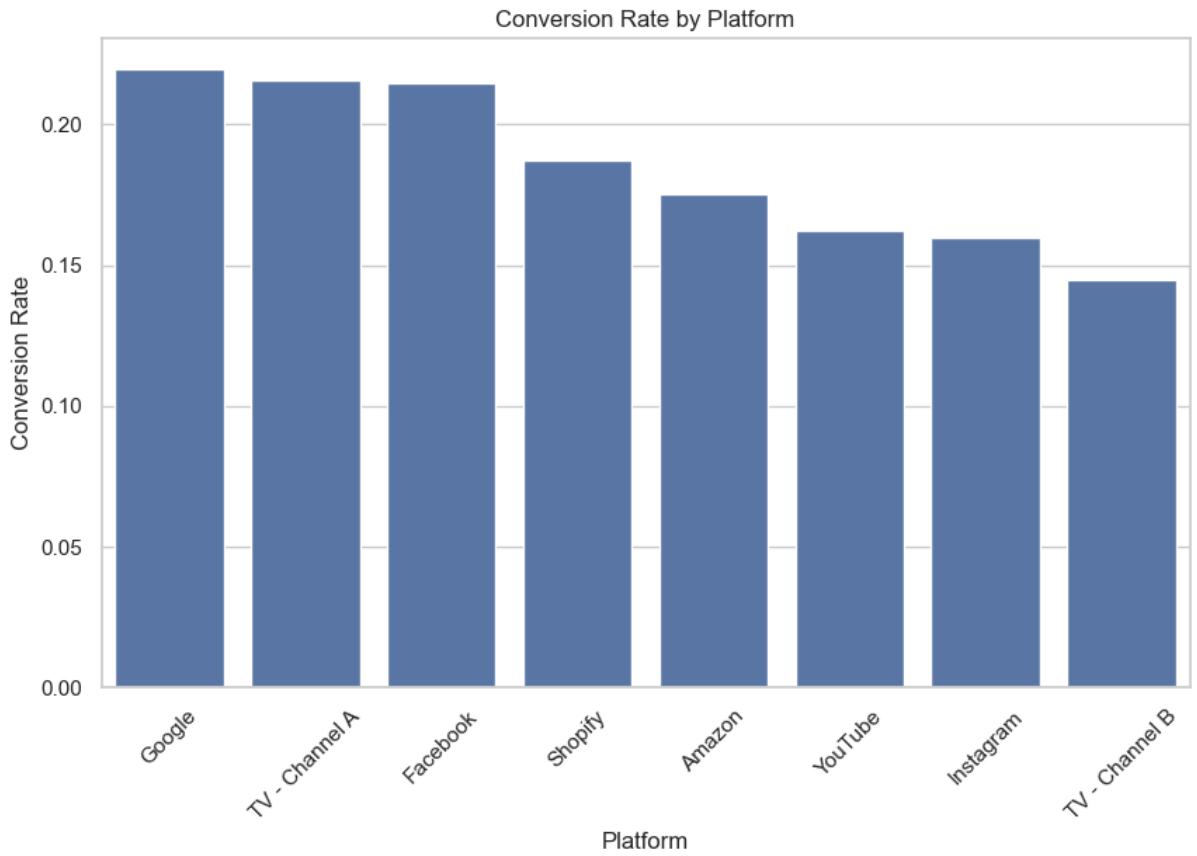


Lowest spend platforms:

- Amazon
- YouTube

This indicates that **traditional TV and search advertising received the majority of the budget allocation**.

Conversion Efficiency



Platforms with **higher conversion efficiency** typically have:

- Higher conversion rates
- Lower CPA

Key observations:

Platform	Spend	Impressions	Clicks	Conversions	CPM	CPV	ConversionRate	ROAS	CPA
Google	766271.820	15516825	733426.0	160981	49.383287	0.049383	0.219492	0.210083	4.760014
TV - Channel A	812134.090	11339940	652907.0	140949	71.617142	0.071617	0.215879	0.173554	5.761900
Facebook	634591.110	12260151	633929.0	136111	51.760464	0.051760	0.214710	0.214486	4.662306
Shopify	544836.380	9814351	601229.0	112475	55.514255	0.055514	0.187075	0.206438	4.844067
Amazon	257284.110	6137278	415085.0	72713	41.921534	0.041922	0.175176	0.282618	3.538351
YouTube	400262.495	8923893	482916.0	78399	44.852902	0.044853	0.162345	0.195869	5.105454
Instagram	553863.260	11287051	664339.0	106144	49.070679	0.049071	0.159774	0.191643	5.218036
TV - Channel B	532137.540	15297641	667445.0	96644	34.785595	0.034786	0.144797	0.181615	5.506162

Top Performing Channels

- Facebook – strong conversion performance relative to spend
- Google – high reach with solid conversions
- Amazon – efficient purchase-focused audience. Best CPA

Moderate Performers

- TV – Channel A (as CPA is high)
- Shopify

These channels generate large traffic volumes but may have slightly higher acquisition costs.

Low Performers

- TV Channel B
- Instagram
- Youtube

These channels generate impressions but comparatively lower conversion efficiency with high CPA.

Audience / Campaign Type Analysis

Campaign types were analysed by aggregating conversions, spend, and clicks.

Insights:

- Some campaign types generate **high engagement (clicks)** but low conversions.
- Others generate **lower traffic but higher conversion efficiency**, indicating stronger purchase intent.

This suggests **different roles for channels in the marketing funnel**:

Funnel Stage	Channels
Awareness	TV, YouTube
Consideration	Facebook, Instagram
Conversion	Google, Amazon

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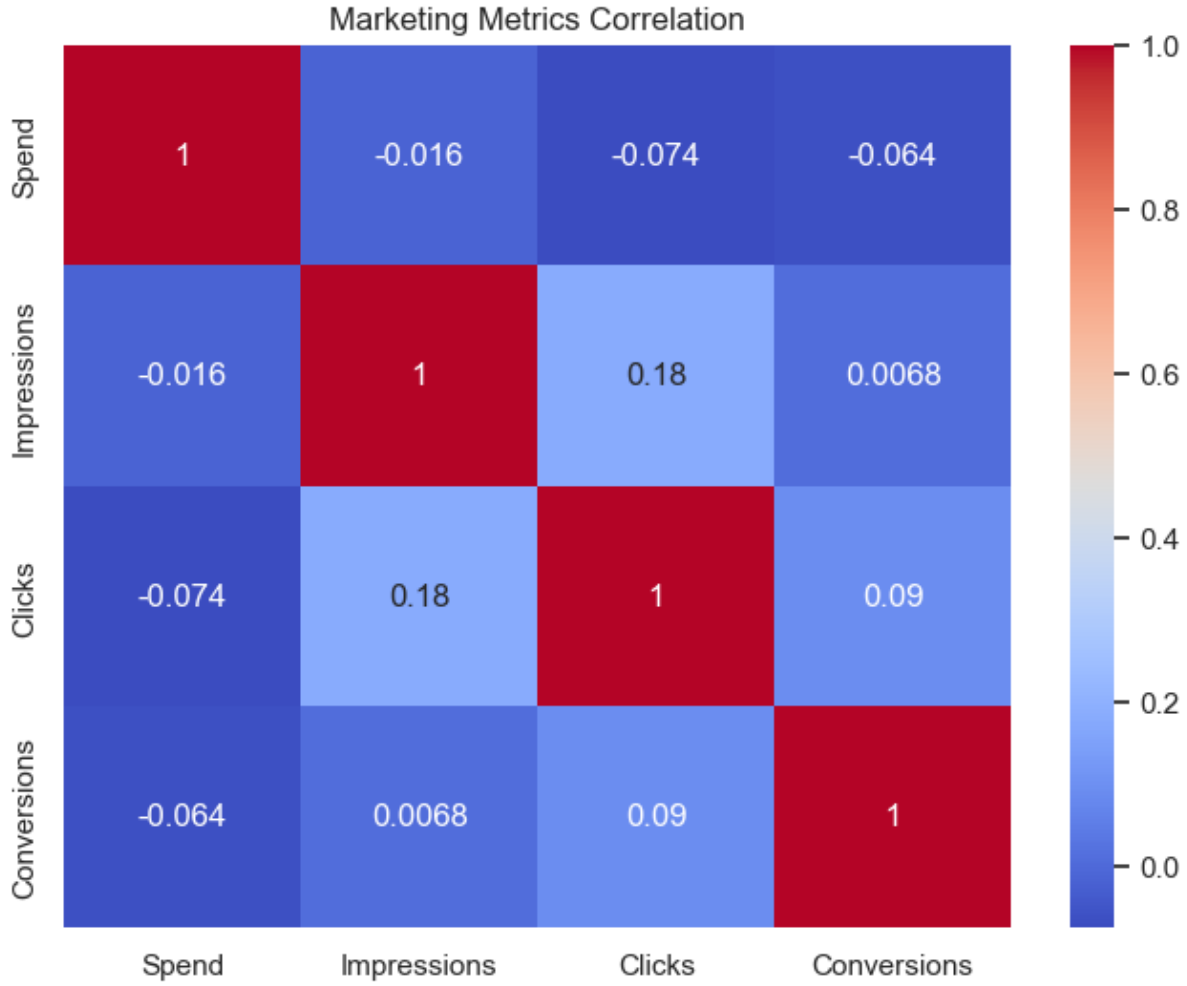
3. Correlation Analysis

A correlation heatmap was generated to analyse relationships between key marketing metrics.

Variables analysed:



- Spend
- Impressions
- Clicks
- Conversions

Key Observations

The correlation values are relatively low:

Relationship	Correlation
Spend vs Impressions	-0.016
Impressions vs Clicks	0.18
Clicks vs Conversions	0.09

Interpretation:

1. **Spend does not strongly correlate with impressions**

This suggests that different platforms have different media pricing structures.

2. **Impressions and clicks have weak correlation**

This indicates that higher reach does not always translate into engagement.

3. **Clicks and conversions have only mild correlation**

This suggests that traffic quality varies across channels.

Implication

The marketing funnel may not be fully optimized:

- Some channels drive impressions but not conversions.
- Audience targeting or landing page optimization may be required.

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4. Visualisation Insights

Spend by Platform

The visualization shows that:

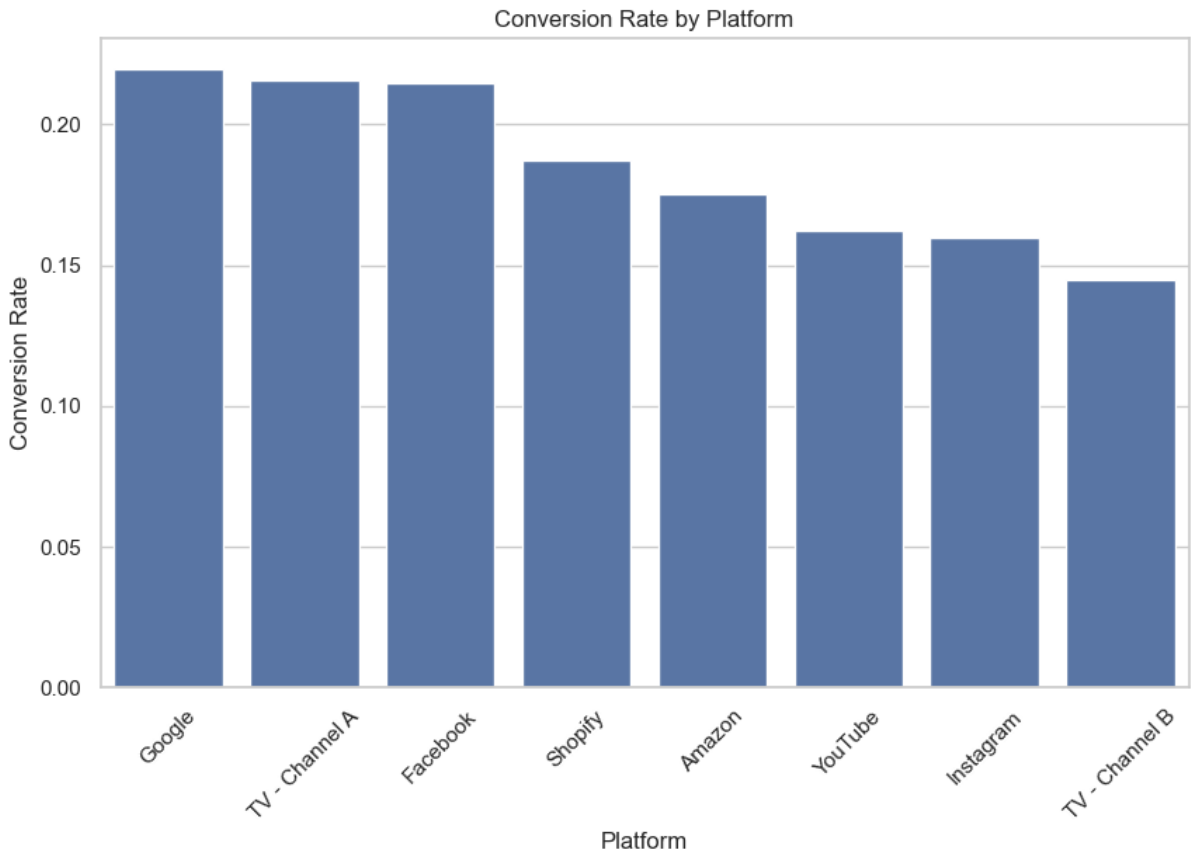
- TV Channel A and Google dominate spending
- Amazon and YouTube receive smaller budget allocations

This suggests the strategy currently focuses heavily on **reach-based channels**.

Conversion Rate by Platform

Conversion rate analysis helps identify **high efficiency channels**.

Channels with stronger conversion performance should receive **incremental budget allocation**.



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5. Optimization Recommendations

Based on KPI analysis and platform performance, the following media mix improvements are recommended.

Increase Budget

Channels with **high conversion efficiency and strong CPA performance**:

- Amazon
- Shopify

These channels appear to generate conversions efficiently.

Maintain Spend

Channels that provide **stable traffic and mid-funnel engagement**:

- Google
- Facebook
- TV Channel A

These channels play an important role in traffic generation and brand visibility.

Reduce Spend

Channels with **lower conversion efficiency**:

- YouTube
- TV Channel B

These platforms may still be valuable for awareness but may require creative or targeting adjustments.

Suggested Media Mix Strategy

Platform	Recommendation
Amazon	Increase
Shopify	Increase
TV Channel A	Maintain
Google	Maintain
Facebook	Maintain
YouTube	Reduce
Instagram	Reduce
TV Channel B	Reduce

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6. Strategic Insights

1. **Budget concentration risk**

Most budget is allocated to a few platforms (TV Channel A and Google). Diversifying towards high-efficiency platforms could improve overall ROI.

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2. **Conversion efficiency opportunity**

Platforms like Amazon demonstrate strong conversion potential and may benefit from additional investment.

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3. **Funnel optimisation**

Channels generating impressions but low conversions may need:

- improved audience targeting
- creative testing
- better landing pages

Conclusion

The analysis highlights significant variation in marketing performance across platforms.

Key takeaways:

- Amazon and Shopify demonstrate strong efficiency potential.
- Google and Facebook remain essential traffic drivers.
- Some awareness channels require optimization to improve conversion efficiency.

By reallocating spend toward higher-efficiency channels and improving funnel optimization, overall campaign ROI can be improved.